

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
df = pd.read_csv('Customer Churn.csv')
df
```

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	\
0	7590-VHVEG	Female	0	Yes	No	1	
1	5575-GNVDE	Male	0	No	No	34	
2	3668-QPYBK	Male	0	No	No	2	
3	7795-CFOCW	Male	0	No	No	45	
4	9237-HQITU	Female	0	No	No	2	
...	...	...	...	...	...	...	
7038	6840-RESVB	Male	0	Yes	Yes	24	
7039	2234-XADUH	Female	0	Yes	Yes	72	
7040	4801-JZAZL	Female	0	Yes	Yes	11	
7041	8361-LTMKD	Male	1	Yes	No	4	
7042	3186-AJIEK	Male	0	No	No	66	

	PhoneService	MultipleLines	InternetService
OnlineSecurity ... \			
0	No	No phone service	DSL
No ...			
1	Yes	No	DSL
Yes ...			
2	Yes	No	DSL
Yes ...			
3	No	No phone service	DSL
Yes ...			
4	Yes	No	Fiber optic
No ...			
...	...	...	...
...			
7038	Yes	Yes	DSL
Yes ...			
7039	Yes	Yes	Fiber optic
No ...			
7040	No	No phone service	DSL
Yes ...			
7041	Yes	Yes	Fiber optic
No ...			
7042	Yes	No	Fiber optic
Yes ...			

	DeviceProtection	TechSupport	StreamingTV	StreamingMovies
Contract \				
0	No	No	No	No
to-month				
1	Yes	No	No	No

One year					
2	No	No	No	No	Month-
to-month					
3	Yes	Yes	No	No	
One year					
4	No	No	No	No	Month-
to-month					
...	...	...	...	...	
...					
7038	Yes	Yes	Yes	Yes	
One year					
7039	Yes	No	Yes	Yes	
One year					
7040	No	No	No	No	Month-
to-month					
7041	No	No	No	No	Month-
to-month					
7042	Yes	Yes	Yes	Yes	
Two year					
	PaperlessBilling		PaymentMethod	MonthlyCharges	
TotalCharges \					
0	Yes		Electronic check	29.85	
29.85					
1	No		Mailed check	56.95	
1889.5					
2	Yes		Mailed check	53.85	
108.15					
3	No	Bank transfer (automatic)		42.30	
1840.75					
4	Yes		Electronic check	70.70	
151.65					
...	...		...	...	
...					
7038	Yes		Mailed check	84.80	
1990.5					
7039	Yes	Credit card (automatic)		103.20	
7362.9					
7040	Yes		Electronic check	29.60	
346.45					
7041	Yes		Mailed check	74.40	
306.6					
7042	Yes	Bank transfer (automatic)		105.65	
6844.5					
	Churn				
0	No				
1	No				
2	Yes				

```

3      No
4      Yes
...    ...
7038   No
7039   No
7040   No
7041   Yes
7042   No

```

```
[7043 rows x 21 columns]
```

```
df.head(5)
```

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure
0	7590-VHVEG	Female	0	Yes	No	1
1	5575-GNVDE	Male	0	No	No	34
2	3668-QPYBK	Male	0	No	No	2
3	7795-CF0CW	Male	0	No	No	45
4	9237-HQITU	Female	0	No	No	2

	MultipleLines	InternetService	OnlineSecurity	...
0	No phone service	DSL	No	...
1	No	DSL	Yes	...
2	No	DSL	Yes	...
3	No phone service	DSL	Yes	...
4	No	Fiber optic	No	...

	TechSupport	StreamingTV	StreamingMovies	Contract
0	No	No	No	Month-to-month
1	No	No	No	One year
2	No	No	No	Month-to-month
3	Yes	No	No	One year

```
No
4      No      No      No      Month-to-month
Yes
```

	PaymentMethod	MonthlyCharges	TotalCharges	Churn
0	Electronic check	29.85	29.85	No
1	Mailed check	56.95	1889.5	No
2	Mailed check	53.85	108.15	Yes
3	Bank transfer (automatic)	42.30	1840.75	No
4	Electronic check	70.70	151.65	Yes

```
[5 rows x 21 columns]
```

```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
#   Column                Non-Null Count  Dtype
---  -
0   customerID            7043 non-null   str
1   gender                 7043 non-null   str
2   SeniorCitizen          7043 non-null   int64
3   Partner                7043 non-null   str
4   Dependents             7043 non-null   str
5   tenure                 7043 non-null   int64
6   PhoneService           7043 non-null   str
7   MultipleLines           7043 non-null   str
8   InternetService        7043 non-null   str
9   OnlineSecurity         7043 non-null   str
10  OnlineBackup           7043 non-null   str
11  DeviceProtection       7043 non-null   str
12  TechSupport            7043 non-null   str
13  StreamingTV            7043 non-null   str
14  StreamingMovies        7043 non-null   str
15  Contract               7043 non-null   str
16  PaperlessBilling       7043 non-null   str
17  PaymentMethod          7043 non-null   str
18  MonthlyCharges         7043 non-null   float64
19  TotalCharges           7043 non-null   str
20  Churn                  7043 non-null   str
```

```
dtypes: float64(1), int64(2), str(18)
```

```
memory usage: 1.9 MB
```

```
df["TotalCharges"] = df["TotalCharges"].replace(" ", "0")
df["TotalCharges"] = df["TotalCharges"].astype("float")
```

```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
```

```
Data columns (total 21 columns):
#      Column      Non-Null Count  Dtype
---  -
0      customerID    7043 non-null    str
1      gender          7043 non-null    str
2      SeniorCitizen    7043 non-null    int64
3      Partner          7043 non-null    str
4      Dependents        7043 non-null    str
5      tenure           7043 non-null    int64
6      PhoneService      7043 non-null    str
7      MultipleLines      7043 non-null    str
8      InternetService    7043 non-null    str
9      OnlineSecurity     7043 non-null    str
10     OnlineBackup        7043 non-null    str
11     DeviceProtection    7043 non-null    str
12     TechSupport         7043 non-null    str
13     StreamingTV         7043 non-null    str
14     StreamingMovies     7043 non-null    str
15     Contract            7043 non-null    str
16     PaperlessBilling    7043 non-null    str
17     PaymentMethod       7043 non-null    str
18     MonthlyCharges      7043 non-null    float64
19     TotalCharges        7043 non-null    float64
20     Churn               7043 non-null    str
```

dtypes: float64(2), int64(2), str(17)

memory usage: 1.8 MB

df.isnull()

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure \
0	False	False	False	False	False	False
1	False	False	False	False	False	False
2	False	False	False	False	False	False
3	False	False	False	False	False	False
4	False	False	False	False	False	False
...	...	...	...	...	...	...
7038	False	False	False	False	False	False
7039	False	False	False	False	False	False
7040	False	False	False	False	False	False
7041	False	False	False	False	False	False

7042	False	False	False	False	False	False
PhoneService MultipleLines InternetService						
OnlineSecurity	...	\				
0	False		False		False	
False	...					
1	False		False		False	
False	...					
2	False		False		False	
False	...					
3	False		False		False	
False	...					
4	False		False		False	
False	...					
...	...		...		...	...
.						
7038	False		False		False	
False	...					
7039	False		False		False	
False	...					
7040	False		False		False	
False	...					
7041	False		False		False	
False	...					
7042	False		False		False	
False	...					
DeviceProtection TechSupport StreamingTV StreamingMovies						
Contract	\					
0	False	False	False	False	False	False
False						
1	False	False	False	False	False	False
False						
2	False	False	False	False	False	False
False						
3	False	False	False	False	False	False
False						
4	False	False	False	False	False	False
False						
...	...	...	...	...	...	...
...						
7038	False	False	False	False	False	False
False						
7039	False	False	False	False	False	False
False						
7040	False	False	False	False	False	False
False						
7041	False	False	False	False	False	False
False						

7042	False	False	False	False
False				
	PaperlessBilling	PaymentMethod	MonthlyCharges	TotalCharges
Churn				
0	False	False	False	False
False				
1	False	False	False	False
False				
2	False	False	False	False
False				
3	False	False	False	False
False				
4	False	False	False	False
False				
...	...	...	...	...
...				
7038	False	False	False	False
False				
7039	False	False	False	False
False				
7040	False	False	False	False
False				
7041	False	False	False	False
False				
7042	False	False	False	False
False				

[7043 rows x 21 columns]

```
df.isnull().sum()
```

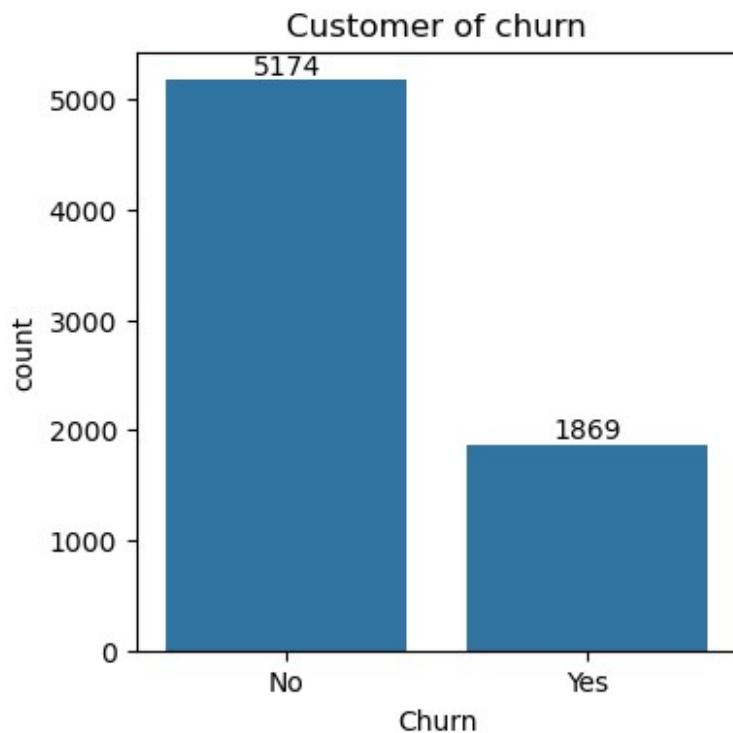
customerID	0
gender	0
SeniorCitizen	0
Partner	0
Dependents	0
tenure	0
PhoneService	0
MultipleLines	0
InternetService	0
OnlineSecurity	0
OnlineBackup	0
DeviceProtection	0
TechSupport	0
StreamingTV	0
StreamingMovies	0
Contract	0
PaperlessBilling	0
PaymentMethod	0

```

MonthlyCharges      0
TotalCharges         0
Churn                0
dtype: int64

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
plt.figure(figsize=(4,4))
ax= sns.countplot(x= 'Churn', data=df)
ax.bar_label(ax.containers[0])
plt.title('Customer of churn')
plt.show()

```

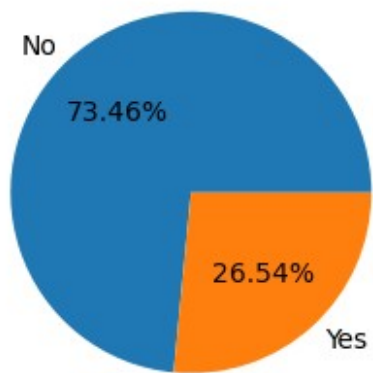


```

import matplotlib.pyplot as plt
plt.figure(figsize=(3,4))
gb = df.groupby("Churn").agg({'Churn': "count"})
plt.pie(gb['Churn'], labels=gb.index, autopct="%1.2f%%")
plt.title("Customer of churn")
plt.show()

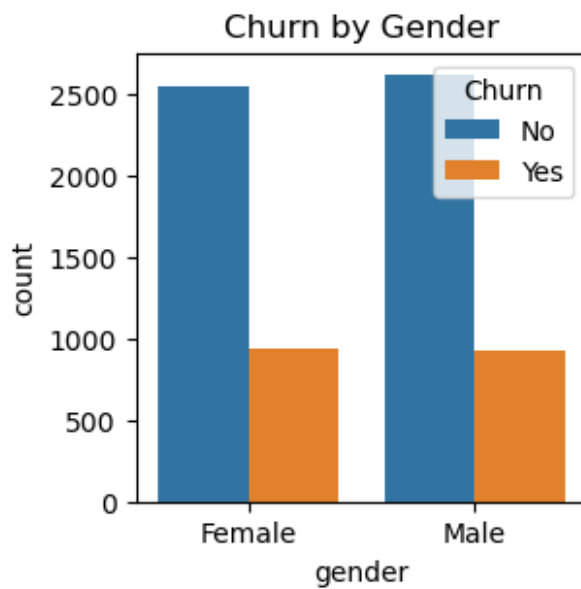
```


Customer of churn

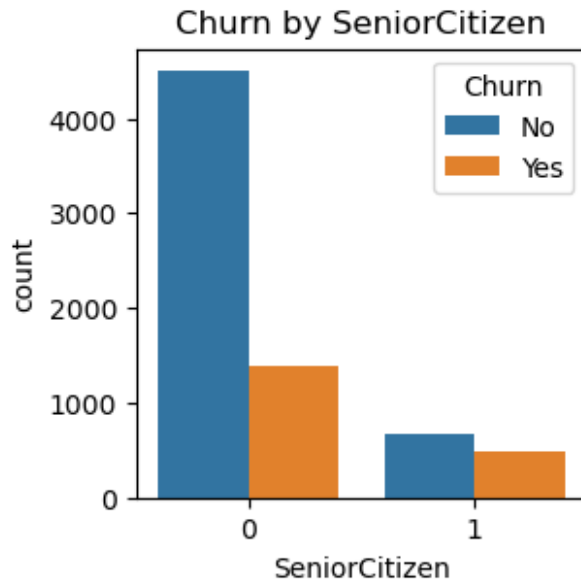


```
plt.figure(figsize=(3,3))
sns.countplot(x="gender", data= df, hue="Churn")
plt.title("Churn by Gender")
plt.show
```

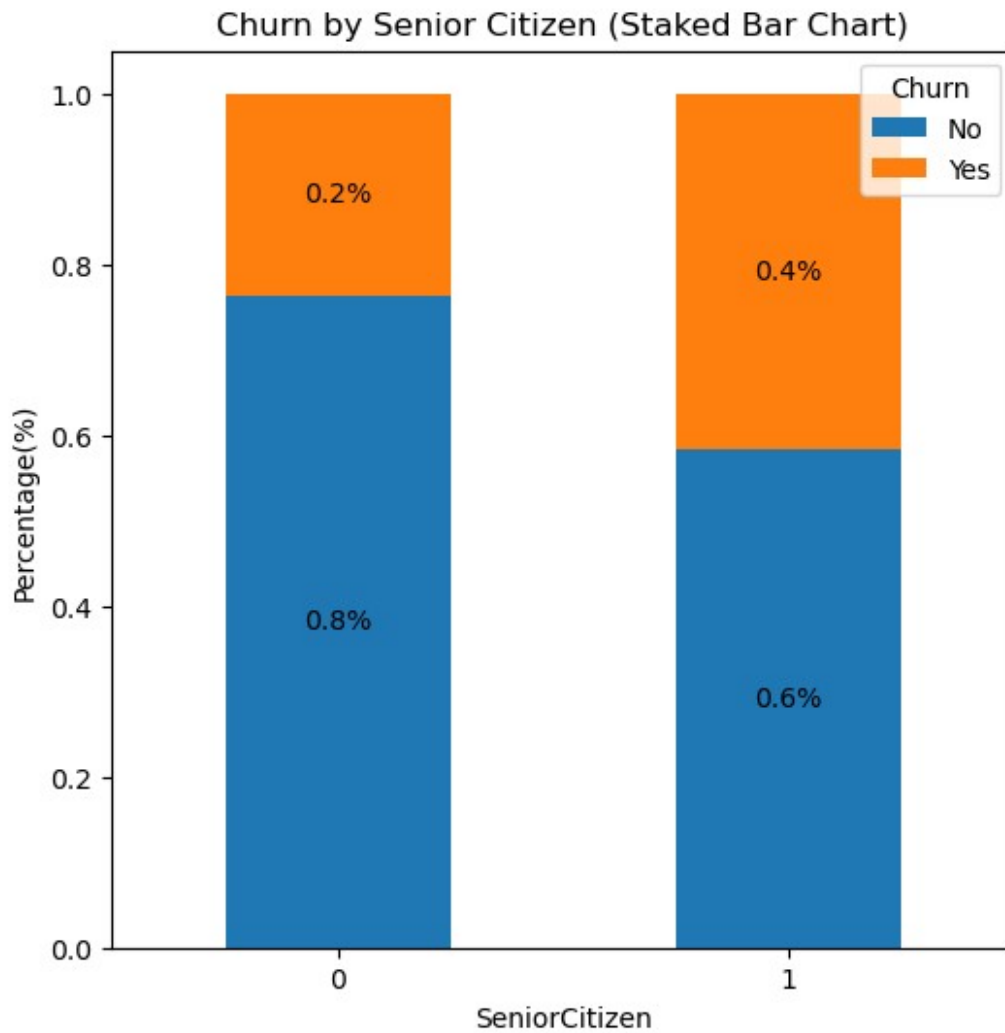
```
<function matplotlib.pyplot.show(close=None, block=None)>
```



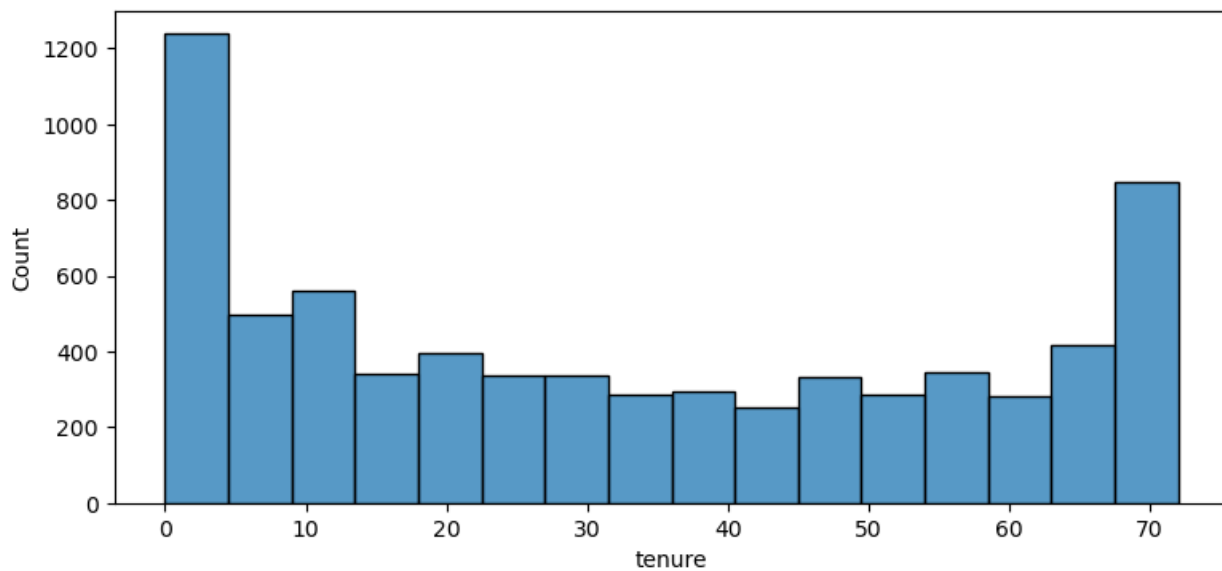
```
plt.figure(figsize=(3,3))
sns.countplot(x="SeniorCitizen", data= df, hue="Churn")
plt.title("Churn by SeniorCitizen")
plt.show()
```



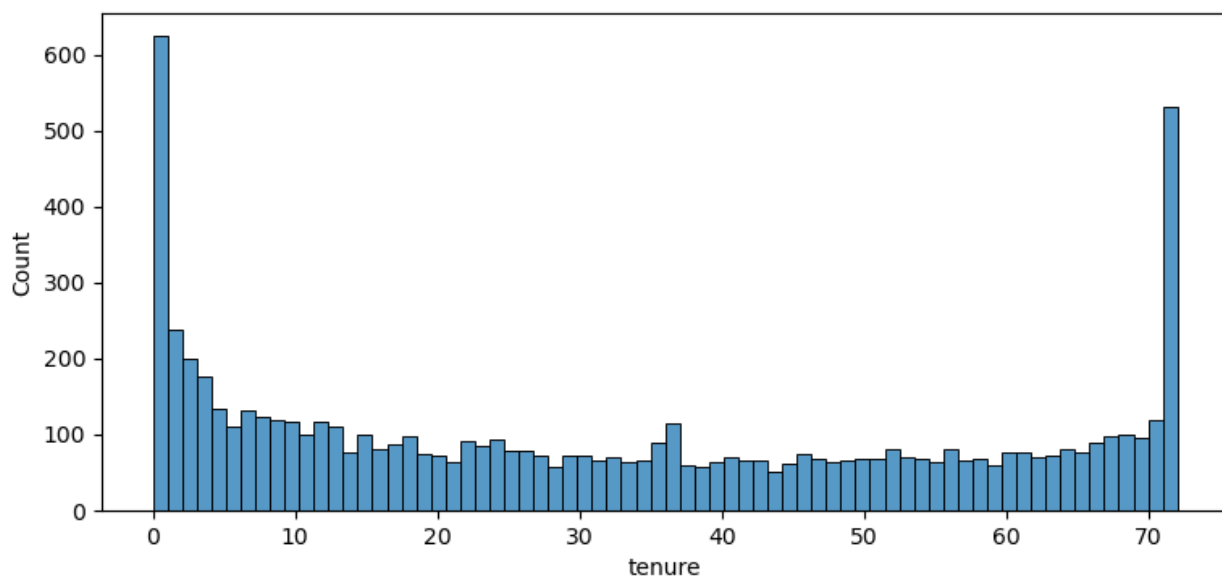
```
total_counts=df.groupby('SeniorCitizen')
['Churn'].value_counts(normalize=True).unstack()
fig, ax= plt.subplots(figsize=(6,6))
total_counts.plot(kind='bar', stacked=True, ax=ax,
color=['#1f77b4', '#ff7f0e'])
for p in ax.patches:
    width, height = p.get_width(), p.get_height()
    x,y = p.get_xy()
    ax.text(x + width / 2, y + height / 2, f'{height:.1f}%',
ha='center', va='center')
plt.title('Churn by Senior Citizen (Stacked Bar Chart)')
plt.xlabel('SeniorCitizen')
plt.ylabel('Percentage(%)')
plt.xticks(rotation=0)
plt.legend(title='Churn', loc='upper right')
plt.show()
```



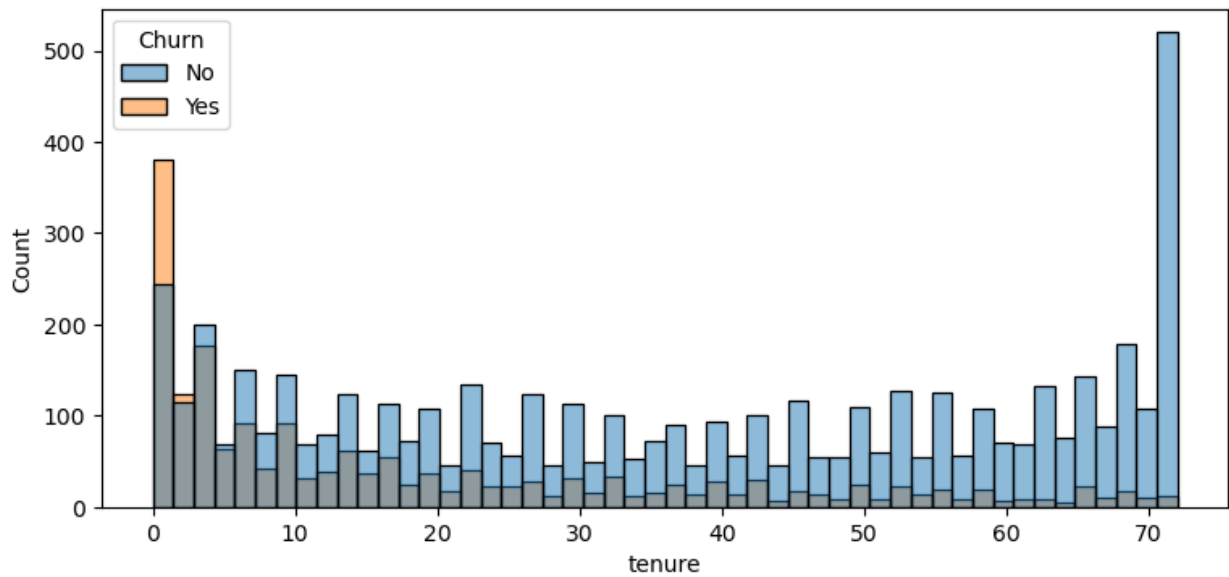
```
plt.figure(figsize = (9,4))  
sns.histplot(x="tenure", data= df)  
plt.show()
```



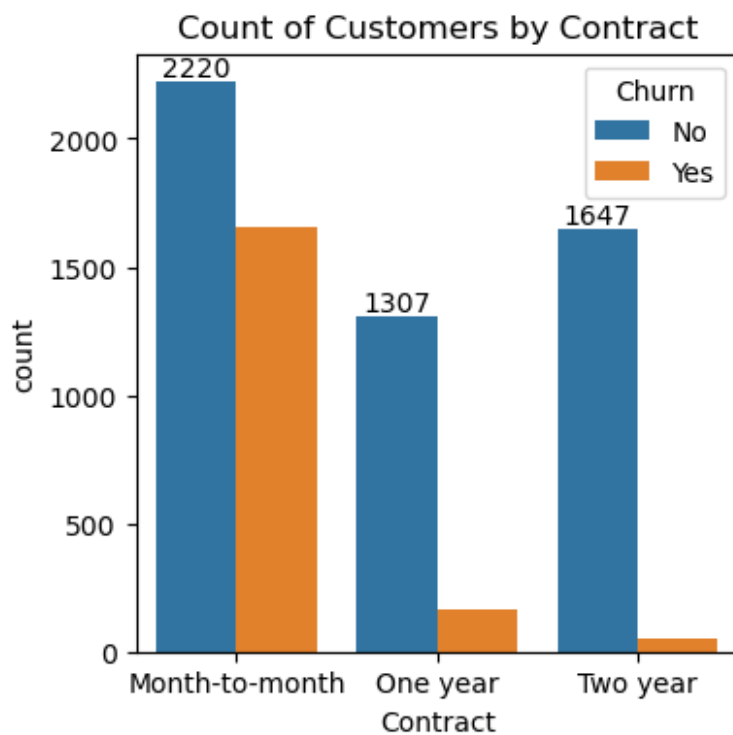
```
plt.figure(figsize = (9,4))
sns.histplot(x="tenure", data= df, bins=70)
plt.show()
```



```
plt.figure(figsize = (9,4))
sns.histplot(x="tenure", data= df, bins=50, hue="Churn")
plt.show()
```



```
plt.figure(figsize=(4,4))
ax= sns.countplot(x= 'Contract', data=df, hue="Churn")
ax.bar_label(ax.containers[0])
plt.title('Count of Customers by Contract')
plt.show()
```



```
columns =
['PhoneService', 'MultipleLines', 'InternetService', 'OnlineSecurity',
```

```
'OnlineBackup', 'DeviceProtection', 'TechSupport', 'StreamingTV', 'StreamingMovies']
```

```
fig, axes = plt.subplots(3,4, figsize=(15,10))
```

```
axes = axes.flatten() # convert 2D axes → 1D list
```

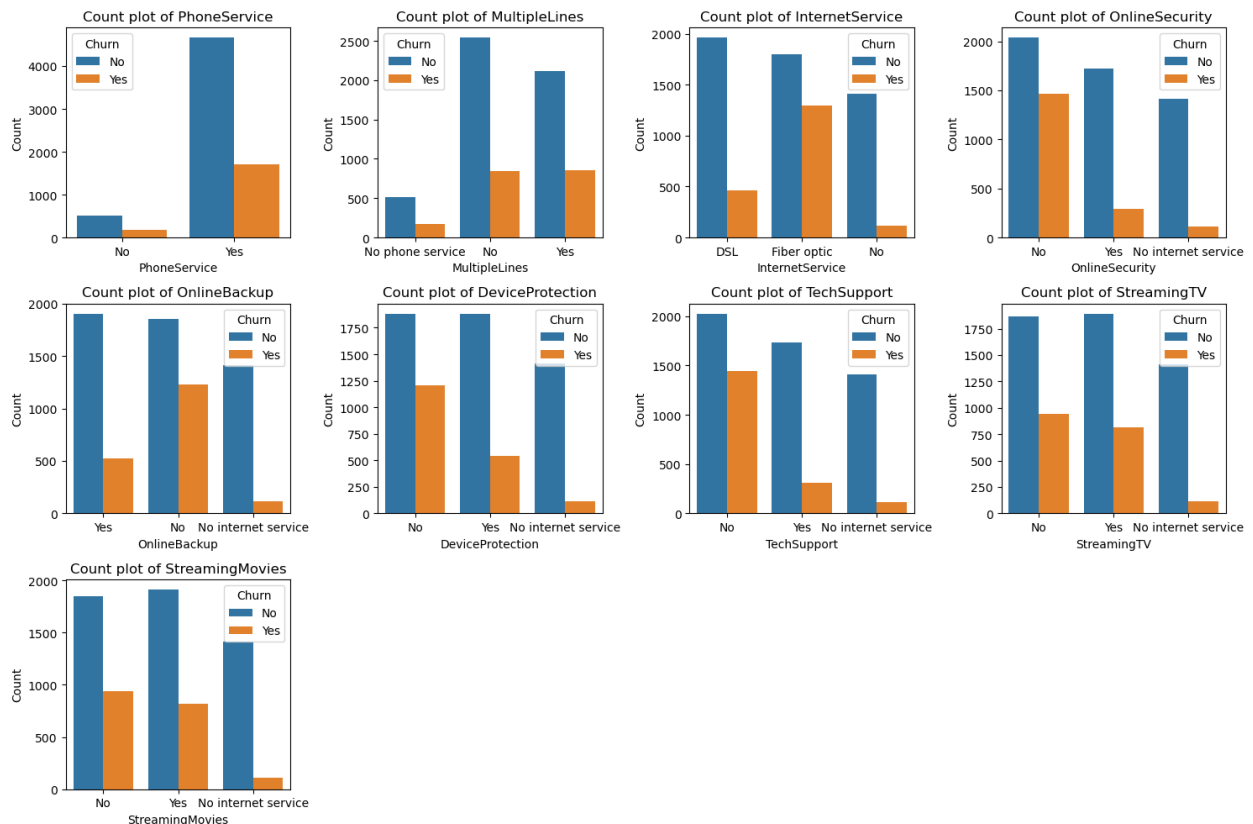
```
for i, col in enumerate(columns):
    sns.countplot(data=df, x=col, ax=axes[i], hue=df["Churn"])
    axes[i].set_title(f'Count plot of {col}')
    axes[i].set_xlabel(col)
    axes[i].set_ylabel('Count')
```

```
# remove extra empty plots
```

```
for j in range(len(columns), len(axes)):
    fig.delaxes(axes[j])
```

```
plt.tight_layout()
```

```
plt.show()
```



```
import matplotlib.pyplot as plt
import seaborn as sns
```

```

columns = ['gender', 'SeniorCitizen', 'Partner', 'Dependents', 'tenure',
           'PhoneService', 'MultipleLines', 'InternetService',
           'OnlineSecurity', 'OnlineBackup']

n_rows = (len(columns) + 3) // 4

fig, axes = plt.subplots(n_rows, 4, figsize=(16, n_rows*4))
axes = axes.flatten()  # ☐ important fix

for i, col in enumerate(columns):
    sns.countplot(data=df, x=col, ax=axes[i])
    axes[i].set_title(f'Count plot of {col}')
    axes[i].set_xlabel(col)
    axes[i].set_ylabel('Count')

# remove extra empty plots
for j in range(len(columns), len(axes)):
    fig.delaxes(axes[j])

plt.tight_layout()
plt.show()

```

